

ADITYA PATEL

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EDUCATION

Bachelor of Science in Computing Science

University of Alberta

Edmonton, AB

Expected Graduation: 2027

Relevant Courses: Algorithms | Machine Learning | Software Engineering | Database Management | Computer Organization & Architecture

EXPERIENCE

Full Stack Developer Intern

Jarillo Consulting

Edmonton, AB

Jan 2026 - April 2026

- Accelerated feature delivery by 30% by serving as Scrum Master for a 7-person Agile team, driving sprint planning, client communications, and rigorous PR reviews.
- Reduced deployment failures by 40% by architecting CI/CD pipelines via GitHub Actions and Vercel on Azure VM infrastructure, streamlining the team's release workflow.
- Shipped Field Notes AI, a cross-platform React Native app serving social workers, by architecting a self-hosted Supabase backend and enforcing SonarQube/CodeQL quality gates that maintained zero critical vulnerabilities across the codebase.

AI Software Developer Intern

JuliGerm

Lesotho, South Africa (Remote)

Apr 2025 - Aug 2025

- Reduced system downtime by 30% by designing and maintaining AI-powered backend pipelines on Microsoft Azure to process agricultural and satellite data at scale.
- Cut farmers' manual tracking effort by 40% by building a custom search engine that unified crop health, soil quality, water usage, and weather data into a single decision dashboard.
- Ensured reliable, scalable delivery of agricultural insights by deploying and maintaining backend services on Microsoft Azure, engineered for fault tolerance across distributed data sources.

AI, Software Developer

MedXAI

Edmonton, AB

July 2024 - August 2025

- Improved hospital resource planning by enabling patient load predictions 2 weeks earlier, achieving 90% forecast accuracy by implementing Python-based generative and classification models on COVID-19 data.
- Reduced researcher reporting time by 25% by building interactive dashboards and analyzing 100+ datasets to visualize real-time and forecasted trends, strengthening model trust and accuracy.
- Increased clinician usability and adoption of AI forecasting tools by collaborating with cross-functional research and clinical teams to integrate prediction models directly into hospital workflows.

ML, Web Development Intern

UAIS (Undergraduate Artificial Intelligence Society)

Edmonton, AB

January 2024 - April 2024

- Deployed NLP pipelines for industrial risk report analysis, achieving 85% prediction accuracy and reducing manual review workload by 40%.
- Improved model prediction reliability by 20% and runtime by 10% through iterative pipeline tuning and optimization.
- Automated report generation with OpenAI API and applied PCA with feature scaling on large datasets, cutting client delivery time by 40%.

PROJECTS

GitImposters: Repo Authenticity Scanner | Next.js, TypeScript, Supabase, Groq API, Tailwind CSS

Team Project | github.com/GitImposters/GitImposters

June 2026

- Flagged inauthentic contributions for hiring teams by building a full-stack tool that analyzes GitHub repositories across 6 engineering dimensions and generates an automated Imposter Score.
- Enabled AI-authored code detection by integrating the Groq API (Llama 3.3 70B) for authorship and commit-pattern analysis, with GitHub OAuth and GitHub App integration for secure repository access.

Field Notes AI | React Native, Supabase, Azure, SonarQube, Jest

Team Project

April 2026

- Maintained zero critical security vulnerabilities across the codebase by enforcing strict quality gates using ESLint, CodeQL, and a comprehensive SonarQube integration for a cross-platform AI app serving social workers.

- Ensured highly reliable and secure delivery by architecting a scalable self-hosted Supabase backend and automating APK build pipelines, cutting manual release effort significantly.

BloodBridge: Blood Donor Platform | React, Next.js, TypeScript, Supabase, PostGIS

HackED 2026

January 2026

- Reduced donor match time by leveraging PostGIS proximity queries and Supabase Realtime push notifications to build a real-time platform connecting hospitals with compatible blood donors.
- Improved match precision for critical cases by designing a geo-aware engine incorporating blood-type compatibility filtering, urgency triage, and live donor availability tracking.

CampusNest: Student Community App | React Native, Expo, Supabase, TypeScript

Personal Project | github.com/campus-nest/campusnest | campusnest.uofacs.ca

April 2026

- Centralized campus community engagement by building a cross-platform mobile app in React Native and Expo, supporting secure user profiles, item listings, and real-time activity feeds.
- Ensured reliable application deployment by integrating Supabase for authentication and data syncing, and establishing automated CI/CD pipelines with Jest test coverage.

Mood Tracker Android App | Java, Android SDK, Firebase, Google Maps SDK

Team Project | Github Repo: github.com/cmput301-w25/project-kernelcrew

April 2025

- Enabled 100+ users to reflect on emotions more consistently by building an Android mood-tracking app with geotagged pins, custom timestamps, and offline sync capabilities.
- Increased daily log completion rates by 25% during testing by implementing authentication, social following, and push notifications in a 6-member Agile team.

BearBazaar: Campus Marketplace | React.js, Express.js, Prisma, SQLite, Node.js, REST API

HackED 2025 | Github Repo: github.com/adityadipakpatel/BearBazaar

February 2025

- Supported secure listings and real-time messaging for 200+ students by building a full-stack marketplace with a React.js frontend and an Express.js + Prisma backend.
- Reduced average API response time by 20% by optimizing SQL queries and REST endpoints, ensuring reliable real-time updates across the platform.

ConsoleTweeter: CLI Social Platform | Python, SQLite, MongoDB

Team Project | Github Repos: [SQLite](#) · [MongoDB](#)

December 2024

- Validated cross-backend correctness on 200+ sample tweets by engineering a CLI social platform with authentication, tweeting, search, and following across both SQL and NoSQL backends.
- Improved search precision and user navigation efficiency by 30% by implementing hashtag parsing and contextual user filters.

MatchHire: Resume Builder | Python, JavaScript, ChatGPT API, PdfLatex API

GHacks 2024

August 2024

- Delivered a fully functional resume auto-generation tool in 24 hours by leading a 4-member team to build a resume parser and LaTeX PDF generator via a web interface.
- Reduced manual formatting time by 80% by integrating LLM APIs with Python and PdfLatex to automate resume structuring and PDF rendering.

RareQuest: Disease Awareness Game | HTML, CSS, JavaScript

Code to Give (Morgan Stanley) 2024 | Github Repo: github.com/adityadipakpatel/RQMO12

May 2024

- Engaged 50+ players in rare disease awareness by creating a browser-based game featuring interactive diagnosis challenges built with HTML, CSS, and JavaScript.
- Improved player retention and cut load time by 20% by optimizing JavaScript game logic and database operations for smoother gameplay performance.

Sign Language Detector | OpenCV, TensorFlow, Python, Jupyter Notebooks

HackED 2024 | Github Repo: github.com/adityadipakpatel/Sign-Language-Detector

January 2024

- Achieved 90% gesture-to-text translation accuracy by building a real-time sign language detection app using TensorFlow and OpenCV.
- Enabled faster digital communication for Deaf and Mute individuals by reducing recognition latency to under 200ms per gesture through model optimization.

EXTRACURRICULAR PROJECTS

CryptoHash: Cryptographic Algorithm (SHA256) Implementation | Python, Cryptography, Cybersecurity

- Deepened applied cryptography knowledge by implementing the SHA256 algorithm from scratch in Python to generate standards-compliant 256-bit hash values.
- Achieved 100% cross-platform reliability by validating full compatibility with Unix/Linux SHA256 utilities through system-level integration tests.

WeatherSage: Weather Data Retriever | Python, OpenWeatherMap API, JSON

- Automated real-time weather retrieval across multiple locations by building a Python tool that fetches, parses, and visualizes data from the OpenWeatherMap API.
- Reduced manual data lookup time by 80% by transforming raw JSON API responses into structured, user-readable outputs.

SecureText: Text Encryption Utility | Python, Cryptography, Security

- Protected sensitive user data by developing a two-passcode encryption and decryption system in Python, ensuring end-to-end privacy.
- Maintained 100% data integrity across all transmission tests by applying cryptographic algorithms to encode text into numeric strings and restore originals securely.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks & Libraries: ReactJS, NodeJS, ExpressJS, Redux, Django, Flask, PyTorch, TensorFlow, scikit-learn

Databases & Cloud: PostgreSQL, MySQL, MongoDB, Microsoft Azure, Amazon Web Services, Google Cloud Platform

Developer Tools: Git/GitHub, Postman, Docker, Android Studio, Linux, Windows